

Threat-Intelligence Kill-Chain Correlation Engine

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Security Operations Centres (SOCs) handle massive volumes of security logs and alerts, making it challenging to identify threats and understand complex attack patterns efficiently. This project proposes a Security Operations and Attack Graph Analysis Engine that applies Data Structures and Algorithms (DSA) to automate threat detection and response. The system scans incoming logs to identify known attack signatures and correlates related alerts to reconstruct multi-step attack sequences, represented as attack kill-chains and graphs. Graph-based analysis is then used to identify critical choke points, which are systems or nodes whose protection can disrupt multiple attack paths. The engine also computes a minimum set of monitoring points required to provide effective coverage across all possible attack paths. Detected alerts are prioritized and routed to appropriate security analysts for further investigation. By combining string matching, hashing, graph traversal, shortest-path analysis, and optimization techniques, the proposed system improves threat visibility, reduces redundant monitoring, and enables faster identification of critical security risks. The project demonstrates how DSA concepts can be applied to solve practical cybersecurity challenges in modern SOC environments.

Signature of the Guide